

Team Name:	Sparkelers
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PROBLEM STATEMENT

! Renewable Energy is Clean... But Not Smart

-  **Solar power is unpredictable** – generation changes every hour.
-  **Batteries drain quickly** due to **poor load planning**.
-  **Sudden demand spikes** cause **overload and system failure**.
-  **No intelligent system to predict or optimize energy usage.**
-  **Result → Power cuts, energy wastage, reduced system life.**

OBJECTIVES

- ⚡ To enable **real-time monitoring** of voltage, current, power, and temperature.
- 🧠 To implement **AI-based load prediction** for smarter energy planning.
- 🔋 To prevent **battery over-discharge and overload conditions**.
- 🔄 To introduce **automatic load prioritization and shedding**.
- 📊 To provide a **live dashboard with analytics and alerts**.
- 🌱 To improve **energy efficiency, system lifespan, and sustainability**.



PROPOSED IOT SOLUTION / APPLICATION

AI-Enabled Smart Microgrid System

-  Real-Time IoT Monitoring of Voltage, Current & Power
-  Cloud-Connected Energy Analytics Platform
-  AI-Based Load Prediction & Smart Optimization
-  Battery Protection & Overload Prevention Mechanism
-  Automatic Load Prioritization & Intelligent Control
-  Live Dashboard with Performance Insights & Alerts
-  Energy-Efficient, Low-Cost & Scalable Solution



REQUIRED TOOLS/SOFTWARE

-  **Arduino IDE** – Firmware Programming
-  **ThingSpeak / Firebase** – Cloud Data Storage
-  **Streamlit / Web Dashboard** – Data Visualization
-  **Python (Scikit-Learn)** – AI Load Prediction Model
-  **GitHub** – Version Control & Deployment

REQUIRED HARDWARE COMPONENTS

- ⚡ **ESP32 Microcontroller** – Core IoT Control Unit
- 🔍 **INA219 Current & Voltage Sensor** – Real-Time Power Monitoring
- 🌡️ **DHT11 Temperature Sensor** – Environmental Condition Tracking
- 🔄 **Relay Module** – Automatic Load Control & Protection
- ☀️ **Solar Panel (10W–20W)** – Renewable Energy Source
- 🔋 **12V Rechargeable Battery** – Energy Storage System
- 💡 **Electrical Load (Bulb / Resistive Load)** – Output Testing
- 🔌 **Breadboard, Wires & Connectors** – Circuit Integration

EXPECTED OUTCOMES

- ⚡ **Real-Time Renewable Energy Monitoring** with accurate system visibility
- 🧠 **AI-Based Load Prediction** for smarter energy planning
- 🔋 **Improved Battery Life** through intelligent power management
- 🔄 Automatic **Overload & Fault Protection**
- 📊 Interactive **Cloud Dashboard** with live analytics
- 🌱 Enhanced **Energy Efficiency & Reduced Power Wastage**
- 🏠 **Scalable Solution** for rural clinics, smart homes & microgrids
- 🚀 Development of Industry-**Ready IoT & Smart Grid Skills**

DECLARATION

- We hereby declare that the idea presented in this project is original and developed solely by our team for this Hackathon.
- This concept has been independently designed, researched, and implemented without copying from any existing solution.
- We take full responsibility for the authenticity, innovation, and technical development of the proposed system.

Team Leader Name & Signature:	Raga Suga S
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